Advanced Machine Learning (AML-MDS)

The project: guidelines

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Please read this document carefully.

The practical work of the course is **one project, carried through the whole course**. You choose one problem now; each block of the course is then applied to that same problem, and a final report puts the pieces together. The point is the comparison: by January you will have seen regularized linear models, kernel methods, neural networks and hybrid or ensemble methods at work on data you know well, and you will be able to say *why* each did what it did. The project is worth 75% of the course grade; the final examination is the remaining 25%. How it is graded is in the separate *rubric*.

1 The project in one table

Deliverable	Weight	What it covers
Proposal	5%	the problem: question, data, loss, baseline, protocol
Milestone 1 — kernel methods	15%	kernel methods on your problem
Milestone 2 — neural methods	15%	neural networks on your problem
Milestone 3 — hybrid and ensemble methods	15%	hybrid and ensemble methods on your problem
Final integrative report	15%	the whole project
Oral presentation and defence	10%	what you did, and why

All dates — when each deliverable is due, when proposals are approved, and from which laboratory your dataset is your cell of the laboratory grid — are in the *course handout*, and only there. All deliveries are made **only through the Racó** (https://raco.fib.upc.edu), by one member of the team.

The project is done in **pairs**: the same pair as in the laboratories, for the whole course. Exceptions (a group of three, a pair that splits) are discussed with me, in advance.

2 Choosing the problem

You have four ways in:

1. **A dataset from a public repository** (Section 7), with a classification or a regression task;
2. **Your own problem** — from your work, a previous course, a research group — provided the data can be used for a course project and shared with me;
3. **A problem taken from a paper** that interests you: you start by reproducing its baseline result, and then carry the problem through the course. You are responsible for getting the data (the same as in the paper, or similar);

4. **A methodological question**: a question about how learning methods behave, rather than about one dataset, that can be asked of *every* family of methods in the course — regularized linear models, kernel methods, neural networks, hybrid and ensemble methods — so that each milestone asks it of the methods of the block just finished. For instance: how they cope with missing values, or with mixed variable types; how their error and their computing cost grow with n or with d; how they react to label noise, to class imbalance, or to a shift between the training and the test distributions; whether their outputs can be read as probabilities; what their regularization actually controls, and how sensitive they are to its tuning.

A methodological project is still an empirical one, with a theoretical or algorithmic core. The question is studied on **synthetic data**, generated so that the property in question can be varied while everything else is known, **and on at least one real dataset** of the kind described below. Everything else is as for the other three ways in: the same proposal, the same three milestones, the same final report, the same results ledger. What changes is where the depth goes. You are expected to explain *why* each family of methods answers the question as it does, and that may mean deriving a property, analysing a special case in closed form, implementing an algorithm yourself rather than calling a library, or proposing a modification of a method and testing it. A question that makes sense for only one family of methods is sent back: the project must cross the whole course, like every other. **The structure is not negotiable**: one milestone per block, each applying that block's methods.
A proposal is a problem, not a choice of method. You do not need to know the methods yet: every block will be applied to the problem you choose now. What you know after Foundations and the first kernel sessions is exactly what a proposal needs.

A good dataset for this course — including the real dataset of a methodological project — is supervised and tabular. Mixed variable types and missing values are welcome: they are part of the course aims. Aim for roughly 1 000 to 10 000 observations and at least ten inputs. Larger is possible, but the kernel methods store an $n \times n$ matrix ($n = 20\,000$ takes 3.2 GB in double precision), so you would subsample for them; make sure you have the computing power the whole course will ask for. Avoid the datasets every tutorial uses (*iris*, *Titanic*, MNIST, ...): there is nothing left to find in them. Projects built on large language models are not recommended.

Choose by the interest the problem raises in you, not by how it "behaves": do not start testing datasets to find an easy one. Some problems are easier in some respects and harder in others, and the choice of problem has little weight in the grade. The same holds for synthetic data, which is a perfectly good choice: its properties are known exactly, which makes comparisons between methods meaningful.

3 The proposal

Two pages at most, as a PDF, with a preliminary title and the names of the pair. Six items, each tied to something you have already seen:

1. **The question**, in one paragraph, and why anyone would want it answered.
2. **The data**: source and licence; n, d and the type of every variable; missing values; the target.
3. **The loss** you will judge predictions by, and why that one: it fixes what the best possible predictor is.
4. **A baseline** you can fit today: a regularized linear or logistic model, with its result.
5. **The protocol**: how the data will be split or resampled, and what is held out until the end (laboratory 0).

6. **One paragraph of expectation**: what a non-linear method might add on this problem, and why you think so.

For a **methodological project** the six items are read as follows: the question is the methodological question, and why its answer matters; the data are the generator of the synthetic data (what it lets you vary, and over what range) and the real dataset; the loss, the baseline and the protocol are given for both; and the expectation says how you think each family of methods will answer the question, and why. Add one or two fundamental references, and one sentence on what you expect to learn. A proposal can come back with a request to change the problem; that is why it is due a week before it must be approved.

4 The milestones and the final report

The milestones are short: **a PDF of three to five pages**, always. Notebooks and scripts go with it as code, not instead of it: the document must stand on its own, without running anything. Each milestone applies the methods of the block just finished to your problem and compares them with *everything you have already done on it* — the baseline, then the earlier milestones. Use the same protocol and the same held-out data throughout, or the comparison means nothing. A milestone answers three questions: what did you fit, and why that; what came out; and what does the difference with the previous models tell you about your problem.

The final integrative report is where the writing effort goes. It is **strictly limited to 12 pages** of main text, references included; appendices go after the main text, in the same PDF, and are read only if needed. It is not the three milestones stapled together: it tells the story of one problem seen through four families of methods, and ends with what you now know about it that you did not know at the start — including the gaps and limitations of your own work. It should not re-explain what was seen in class, and it should contain no table or figure that the text does not discuss.

The oral presentation is short, and most of it is discussion. You will be asked about your choices and your code: *code you did not write is code you cannot defend*, and "that is what the assistant gave me" is not an answer.

5 What every delivery contains

Every document is written with the course template (`aml-project-template.zip` in the Racó; it works in Overleaf too). It fixes the layout on which the page limits are counted, prints under the title how many pages your main text takes, and keeps two things identical across your deliverables: the **project card**, six lines that state your problem, and the **results ledger**, one row for every model you fit during the course. You add rows to the ledger and never delete one; each milestone prints it whole, so the comparison with everything you did before is always in one place. A document that does not use the template is returned.

One compressed file, through the Racó, with:

1. the **document**, as a PDF written with the template, with its `.tex` sources;
2. **all the code** needed for the results (as files, not as links to a repository), in R or Python; pre-processing or plotting may be done in any language, and its code is included too;
3. the **data**, or links to it if it is large or its licence forbids copying;
4. a plain-text **README** with the exact steps to reproduce every number in the document: seeds set once at the top, package versions recorded, nothing edited by hand.

Theoretical rigour applies to all of it. Whether your project starts from a dataset or from a methodological question, you are expected to state your assumptions, use precise notation,

justify every choice, and explain why a method works or fails on your problem — not merely report that it did. The conditions for a high grade are, in this order:

1. the proper use of the methods of the corresponding block;
2. theoretical rigour and understanding;
3. care in obtaining the results: protocol, metrics, uncertainty, experimental design;
4. the quality of the results: generalization error, simplicity, interpretability;
5. the quality of the writing: conciseness, completeness, clarity.

Sources and AI assistants. Cite everything you use: papers, code, data, tools. If you use an AI assistant, say where, every time, as explained in the course handout, section *Studying with an AI assistant*. An unacknowledged use counts as plagiarism.

6 Pre-processing

Each problem needs its own cleaning and preparation, and it can take a good part of your time: it has a large effect on everything that follows. Look at the data carefully before doing anything, and attend to (not necessarily in this order): missing values; outliers; incoherent or incorrect values; irrelevant variables; redundant variables; the coding of nominal and binary variables; new variables worth constructing; normalization (e.g. standardization); and transformations (e.g. for skewness). **Every pre-processing step that learns something from the data — a mean, a scale, an imputation model — is learned on the training part only**, inside the resampling loop. Otherwise the test data leak into the model and the error you report is optimistic.

7 Where to find data

- OpenML: https://www.openml.org/search?type=data
- UCI Machine Learning Repository: https://archive.ics.uci.edu/
- StatLib: http://lib.stat.cmu.edu/datasets/
- Delve: https://www.cs.toronto.edu/~delve/data/datasets.html
- University of Edinburgh, suggested datasets: https://www.inf.ed.ac.uk/teaching/courses/irds/miniproject-datasets.html
- Luís Torgo's regression datasets: https://www.dcc.fc.up.pt/~ltorgo/Regression/DataSets.html
- Kaggle: https://www.kaggle.com/datasets. Check the licence, and keep in mind that many of its datasets have been pre-processed, often more than once, and are no longer the original data.
- Public open-data portals (government, health, environment), which often hold problems nobody has modelled yet.